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TASK: EBS Volume Management

- Create a new EBS volume and attach it to an EC2 instance.
- Format and mount it.
- Make it persistent across reboots.

1. Objective

The objective of this lab is to:

- Create a new Amazon EBS volume.
- Attach it to an EC2 instance.
- Format and mount the volume.
- Make the mount persistent across reboots.
- Verify functionality.

2. Steps Performed

Step 1: Launch an EC2 Instance

- Launched a new EC2 instance (**MyLabInstance**) in the **us-east-1a** Availability Zone.
- Instance type: t2.micro (Free Tier eligible).
- Verified it was running.

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page is active, displaying a table with one instance: 'MyLabInstance' (ID: i-0d2efdcf1cc583281), which is in the 'Running' state. The instance is a 't2.micro' type, located in the 'us-east-1a' Availability Zone. Below the table, the details for the selected instance are shown, including the 'Instance summary' section.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
MyLabInstance	i-0d2efdcf1cc583281	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-3-82-9-18.compute...	3.82.9.18

i-0d2efdcf1cc583281 (MyLabInstance)

Instance summary

Instance ID: i-0d2efdcf1cc583281

Public IPv4 address: ec2-3-82-9-18.compute-1.amazonaws.com

Private IPv4 addresses: 172.31.16.128

Step 2: Create an EBS Volume

- Created an **8 GiB General Purpose SSD (gp3)** volume in the same Availability Zone (**us-east-1a**).
 - Attached it to the instance as **/dev/sdf** (appears as **/dev/xvdf** in the OS).
- Choose your instance (MyLabInstance) → Device name: /dev/sdf (AWS may map it to /dev/xvdf) → Attach.**

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Volumes:

Successfully created volume vol-0895efa256991e2c5.

Volumes (1/2) Info

Volume ID: vol-0895efa256991e2c5

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status
vol-0895efa256991e2c5	vol-0895efa256991e2c5	gp3	8 GiB	3000	125	-	2025/09/04 16:27 GMT+5:30	us-east-1a	Available	No alarms
vol-096e8a1612d03a54b	vol-096e8a1612d03a54b	gp3	8 GiB	3000	125	snap-0f9367e...	2025/09/04 16:25 GMT+5:30	us-east-1a	In-use	No alarms

Volume ID: vol-0895efa256991e2c5

Volume ID: vol-0895efa256991e2c5

Size: 8 GiB

Type: gp3

IOPS: 3000

Throughput: 125

Status check: Okay

Volume state: Available

Multi-Attach enabled: No

Created: Thu Sep 04 2025 16:27:21 GMT+0530 (India Standard Time)

Managed: false

Operator: -

Fast snapshot restored: No

Availability Zone: us-east-1a

Outposts ARN: -

Attached resources: -

Source: Snapshot ID: -

Encryption: Not encrypted

KMS key ID: -

KMS key alias: -

KMS key ARN: -

Select the new volume → Actions → Attach Volume.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Volumes:

Successfully created volume vol-0895efa256991e2c5.

Volumes (1/2) Info

Volume ID: vol-0895efa256991e2c5

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Created	Availability Zone	Volume state	Alarm status
vol-0895efa256991e2c5	vol-0895efa256991e2c5	gp3	8 GiB	3000	125	-	2025/09/04 16:27 GMT+5:30	us-east-1a	Available	No alarms
vol-096e8a1612d03a54b	vol-096e8a1612d03a54b	gp3	8 GiB	3000	125	snap-0f9367e...	2025/09/04 16:25 GMT+5:30	us-east-1a	In-use	No alarms

Volume ID: vol-0895efa256991e2c5

Volume ID: vol-0895efa256991e2c5

Size: 8 GiB

Type: gp3

IOPS: 3000

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Status check: Okay

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Operator: -

Fast snapshot restored: No

Availability Zone: us-east-1a

Outposts ARN: -

Attached resources: -

Source: Snapshot ID: -

Encryption: Not encrypted

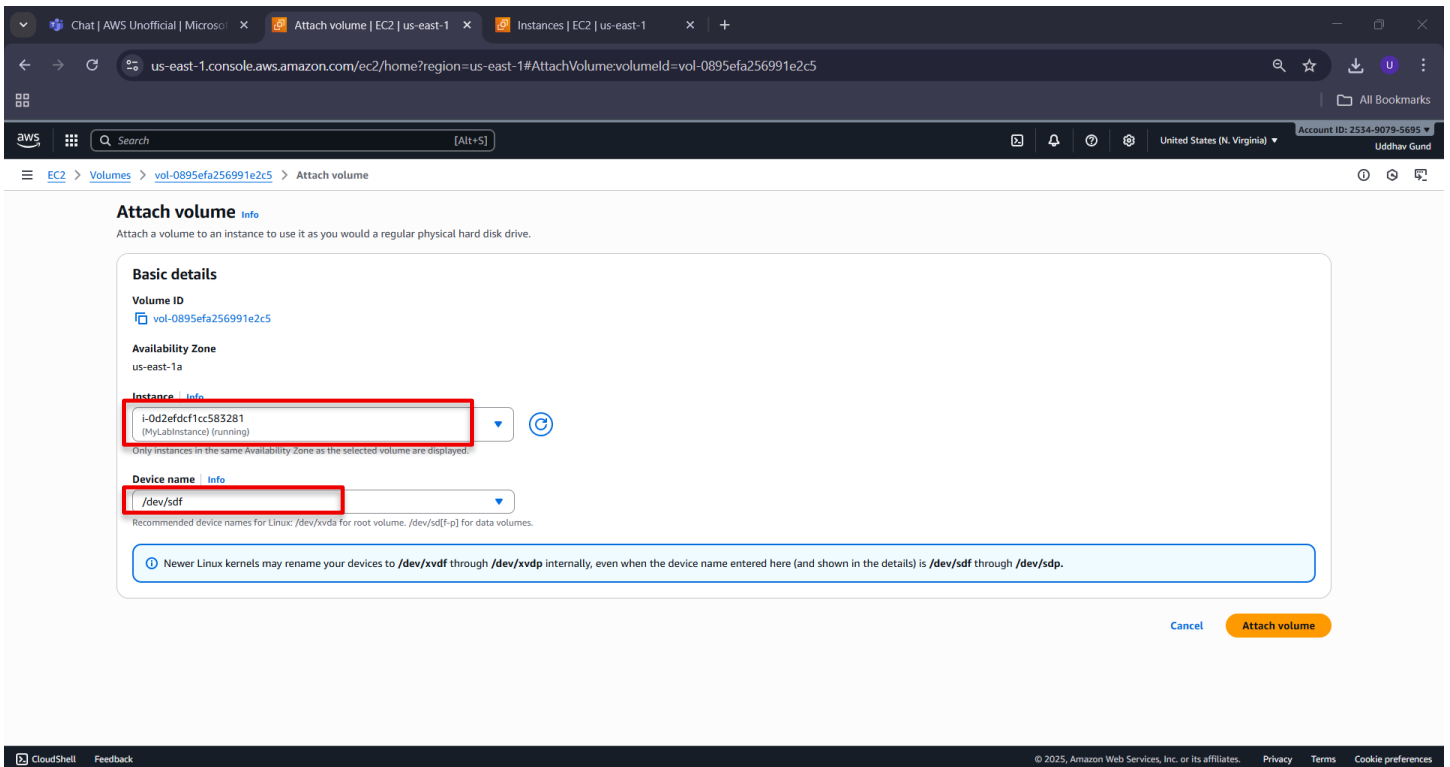
KMS key ID: -

KMS key alias: -

KMS key ARN: -

Actions

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Manage tags
- Fault injection



Step 3: Connect to Instance and Verify

- Used EC2 Instance Connect to connect.
- Ran `lsblk` to verify the new disk was attached.

Step 4: Format and Mount the Volume

Commands used:

Format:

```
sudo mkfs -t ext4 /dev/xvdf
```

Create mount directory:

```
sudo mkdir /mnt/data
```

Mount volume:

```
sudo mount /dev/xvdf /mnt/data
```

Verify:

```
df -h /mnt/data
```

- Verified that the new filesystem was available at `/mnt/data`.

Step 5: Make Mount Persistent

To ensure that the EBS volume automatically mounts even after the instance is rebooted, the following steps were performed:

- Found the UUID using(`blkid` command to get the unique identifier of the device):

```
sudo blkid /dev/xvdf
```

Example output:

```
/dev/xvdf: UUID="f8c7ef35-e8fa-43bf-8f5e-c60d2a228d01" BLOCK_SIZE="4096" TYPE="ext4"
```

- Edited `/etc/fstab` and added:

Opened the `/etc/fstab` file using:

```
sudo nano /etc/fstab
```

Added the following entry at the bottom:

```
UUID=f8c7ef35-e8fa-43bf-8f5e-c60d2a228d01    /mnt/data
```

- UUID ensures the correct disk is mounted.
- /mnt/data is the mount point.
- ext4 is the filesystem type.
- defaults,nofail allows boot even if the volume is missing.
- 0 2 specifies fsck check order.

• Verified with:

```
sudo mount -a
df -h /mnt/data
```

• Rebooted the instance and confirmed /mnt/data was mounted automatically.

Rebooted the instance with:

```
sudo reboot
```

• After reconnecting, ran:

```
df -h /mnt/data
```

Output confirmed /dev/xvdf was automatically mounted at /mnt/data.

[NOTE: EXECUTION STEPS SCREENSHOT OF STEP 3.4.5]

The screenshot displays the AWS Management Console interface for connecting to an EC2 instance. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ConnectToInstance:instanceId=i-0d2efdcf1cc583281`. The console header includes the AWS logo, a search bar, and navigation links for EC2, Instances, and the specific instance `i-0d2efdcf1cc583281`.

The main content area is titled "Connect" and includes an "Info" link. Below the title, it states "Connect to an instance using the browser-based client." There are four tabs: "EC2 Instance Connect" (selected), "Session Manager", "SSH client", and "EC2 serial console".

The "EC2 Instance Connect" tab shows the following details:

- Instance ID:** `i-0d2efdcf1cc583281` (MyLabInstance)
- Connection type:** Two radio buttons are present. "Connect using a Public IP" is selected, with the subtext "Connect using a public IPv4 or IPv6 address". The other option is "Connect using a Private IP" with the subtext "Connect using a private IP address and a VPC endpoint".
- Public IPv4 address:** `3.82.9.18`
- IPv6 address:** (Not specified)
- Username:** `ec2-user`

A note at the bottom of the form states: "Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username." At the bottom right, there are "Cancel" and "Connect" buttons.

The footer of the console shows "CloudShell" and "Feedback" links on the left, and copyright information "© 2025, Amazon Web Services, Inc. or its affiliates." along with "Privacy", "Terms", and "Cookie preferences" links on the right.

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0d2efdcf1cc583281&osUser=ec2-user®ion=us-east-1&sshP...

```
[ec2-user@ip-172-31-38-250 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINT
xvda        202:0    0   8G  0 disk 
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0   1M  0 part 
├─xvda128   259:1    0  10M  0 part /boot/efi
└─xvdf      202:80    0   8G  0 disk 
[ec2-user@ip-172-31-38-250 ~]$ sudo mkfs -t ext4 /dev/xvdf
mkfs2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2097152 4k blocks and 524288 inodes
Filesystem UUID: f8c7ef35-e8fa-43bf-8f5e-c60d2a228d01
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-172-31-38-250 ~]$ sudo mkdir /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ sudo mount /dev/xvdf /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ df -h /mnt/data
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvdf        7.8G   24K   7.4G   1% /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ sudo blkid /dev/xvdf
/dev/xvdf: UUID="f8c7ef35-e8fa-43bf-8f5e-c60d2a228d01" BLOCK_SIZE="4096" TYPE="ext4"
[ec2-user@ip-172-31-38-250 ~]$ sudo nano /etc/fstab
```

i-0d2efdcf1cc583281 (MyLabInstance)

PublicIPs: 3.82.9.18 PrivateIPs: 172.31.38.250

CloudShell Feedback

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us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0d2efdcf1cc583281&osUser=ec2-user®ion=us-east-1&sshP...

```
[ec2-user@ip-172-31-38-250 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINT
xvda        202:0    0   8G  0 disk 
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0   1M  0 part 
├─xvda128   259:1    0  10M  0 part /boot/efi
└─xvdf      202:80    0   8G  0 disk 
[ec2-user@ip-172-31-38-250 ~]$ sudo mkfs -t ext4 /dev/xvdf
mkfs2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2097152 4k blocks and 524288 inodes
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Superblock backups stored on blocks:
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Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-172-31-38-250 ~]$ sudo mkdir /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ sudo mount /dev/xvdf /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ df -h /mnt/data
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvdf        7.8G   24K   7.4G   1% /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ sudo blkid /dev/xvdf
/dev/xvdf: UUID="f8c7ef35-e8fa-43bf-8f5e-c60d2a228d01" BLOCK_SIZE="4096" TYPE="ext4"
[ec2-user@ip-172-31-38-250 ~]$ sudo nano /etc/fstab
[ec2-user@ip-172-31-38-250 ~]$ sudo mount -a
[ec2-user@ip-172-31-38-250 ~]$ df -h /mnt/data
Filesystem      Size  Used Avail Use% Mounted on
/dev/xvdf        7.8G   24K   7.4G   1% /mnt/data
[ec2-user@ip-172-31-38-250 ~]$ sudo reboot

Broadcast message from root@localhost on pts/1 (Thu 2025-09-04 11:18:18 UTC):

The system will reboot now!

[ec2-user@ip-172-31-38-250 ~]$
```

i-0d2efdcf1cc583281 (MyLabInstance)

PublicIPs: 3.82.9.18 PrivateIPs: 172.31.38.250

CloudShell Feedback

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us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0d2efdcf1cc583281&osUser=ec2-user&sshPort=22&addressFam... United States (N. Virginia) Account ID: 2534-9079-5695 Uddhav Gund

i-0d2efdcf1cc583281 (MyLabInstance)
PublicIPs: 3.82.9.18 PrivateIPs: 172.31.38.250

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- Successfully created, attached, formatted, and mounted an EBS volume.
- Configured persistence across reboots using `/etc/fstab`.
- Verified functionality.